

An Agent-Oriented Framework to Support Rapid Early-Stage Object-Oriented Design

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Abstract

Object-Oriented (OO) design, as a cornerstone methodology in software engineering, has been playing a pivotal role in software development, during which OO design models are often crafted. However, the challenge is always on intensive cognitive effort and expertise required for modeling. In the past, automatically generating models from requirements has been considered a promising direction to overcome this challenge. The model-driven engineering (MDE) community has intensively made effort towards this direction. Though some progresses have been made with the help of natural language processing (NLP) techniques, fully automated solutions are rare due to the limitation of NLP techniques themselves, and generated models are not full correct and/or complete. Towards this direction, in this paper, we introduce an agent-oriented methodology with tool support, named RapidOOD, which leverages large language models (LLMs) for the generation, quality assessment, and refactoring of OO design models, by taking requirements models as the input. Our evaluation of RapidOOD with six requirement models demonstrates that automatically generated and subsequently optimized design models achieve 89.20% (without human) and 94.56% (human-in-the-loop collaboration) of expert-level quality metrics, closely matching models crafted manually by experts. Furthermore, our approach achieves 1.39 times the efficiency of the manual method.

Keywords: Object-Oriented, Agent-Oriented, Model Generation, Model Refactoring, Large Language Model

1 Introduction

Software design is a fundamental part of software development, covering two primary stages: architectural design and detailed design[1]. The object-oriented design method, which facilitates the creation of different abstract types during the design process, is more intuitive than the process-oriented design method and has garnered considerable attention. After decades of evolution, object-oriented design has become one of the main approaches to software design and development[2]. Utilizing models is a vital supplementary technique in software design. Since textual descriptions are challenging to employ for development, review, and maintenance, abstraction through graphical modeling languages is used to enhance communication between customers and developers, support requirements validation, facilitate architectural and detailed design, aid in prototyping, enable code generation, and more. With Unified Modeling Language(UML), requirements can be modeled as a lightweight formal requirements model that consists of a use case diagram, a conceptual class diagram, use case definitions specified by system sequence diagrams, and the contracts of their system operations[3–5]. A system operation contract is formally specified by a pair of pre- and post-conditions in Object Constraint Language (OCL). Designs can be modeled using class diagrams and sequence diagrams. Works in requirements phase like RM2PT[6–9] provides features for requirements modeling, analysis, and automated prototype generation from requirements models. Users can rapidly validate and evolve functional requirements through RM2PT to produce high-quality validated requirements models. The validated requirements model can be used as input to the design phase.

Object-oriented design in software development is a knowledge and effort-intensive activity. Designers must invest significant time and effort to create an appropriate design solution for given requirements. Different scenarios and requirements necessitate different design solutions and trade-offs, often depending on the designer’s experience and subjective judgment. Additionally, evaluating the design quality early in the project is challenging and usually not feasible.

This paper aims to reduce the time and effort required during the software design phase through automated methods, lessen the dependence on the designer’s expertise, and provide a quantifiable and comprehensible way to reflect design quality early in the project. The main difficulties and challenges in generating design from requirements and subsequently evaluating and optimizing the initial design include:

1. **Requirements capture stakeholders’ needs, goals, and desired system behaviors. In our context, we focus on functional requirements specified through lightweight formal models,** while design describes system’s internal implementation. Design is a concretization and refinement of requirements, containing more detailed information. It is challenging to supplement the missing information needed to transition from requirements to design.
2. Contracts describe the system state constraints before and after executing operations. Key challenges include determining whether there is a mapping from system states to system implementations (combinations of OO basic operations) and how to map these into operation sequences automatically.

75 3. In practice, design often requires multiple iterations of quality assessment and refac-
76 toring. A critical challenge is adopting suitable iteration strategies and methods to
77 ensure iterations progress toward improved design.

78 These challenges highlight the complexity of software design, underscoring the
79 potential benefits of automating certain aspects of the design process to improve
80 efficiency, quality, and adaptability to changing technological trends.

81 This paper presents an agent-oriented methodology named RapidOOD. It supports
82 the rapid generation, quality assessment, and refactoring of object-oriented designs.
83 This methodology integrates model-driven rule-based agents and LLMs Agents.
84 RapidOOD takes lightweight formal requirement models[10] as input, automatically
85 generates initial object-oriented design models, and computes design quality metrics
86 based on the model elements. It can offer simple rule-based refactoring suggestions
87 based on these quality metrics, and it is also capable of providing complex refactoring
88 recommendations based on LLMs. Designers can iterate on the design model regarding
89 the quality metrics and refactoring suggestions, where each modification’s impact on
90 design quality is immediately reflected through the metrics. A relatively high-quality
91 design can be quickly achieved by automating generation, quality assessment, and
92 refactoring suggestions.

93 We have developed a tool to automate the method. This method was evaluated
94 using six requirement models. In terms of generating design models, the main focus
95 was on assessing the coverage of the proposed rules, with the quality assessment results
96 represented by the ratio of successfully generated class diagram elements and sequence
97 diagram elements. The quality of the automatically generated design models and the
98 quality of the design models after a few iterations were evaluated against the quality
99 of models manually constructed by experts. The time savings achieved by adopting
100 this method were assessed on the six cases. Overall, the experimental results were
101 satisfactory.

102 The contributions of the paper are as follows:

- 103 1. **An Agent-Oriented Approach that Integrates Model-Driven Rule-Based**
104 **Agents and LLM Agents:** This paper presents an agent-oriented methodology
105 which facilitates the rapid generation, assessment, and refactoring of object-oriented
106 designs. This methodology combines model-driven rule-based agents and LLM
107 agents. The software design process is partitioned into an initial design phase fea-
108 turing low degrees of freedom, an quality assessment phase, and an refactoring
109 phase with high degrees of freedom. During the phases with low degrees of freedom,
110 rule-based Agents are utilized to avoid problems like hallucinations resulting from
111 the premature application of LLMs. In the phases with high degrees of freedom,
112 LLMs are incorporated to fully exploit their knowledge-emergence capabilities.
- 113 2. **A Method for Analyzing the Executability of OCL Contracts and Path**
114 **Reasoning:** Our work introduces an automated approach for analyzing the exe-
115 cutability of a given OCL contract and determine a reasonable implementation
116 path. This method effectively bridges the gap between requirements and design,
117 providing an efficient, automated pathway for object-oriented software design.

118 **3. Design refactoring with Embedding quality assessment Results in LLM**
119 **Prompts:** Utilizing a model-driven methodology, this research advances design
120 quality quality assessment from code-based assessments to an innovative model-
121 based framework. The design quality quality assessment results obtained through
122 the model-driven method are embedded into LLM Prompts to guide the LLM to
123 propose corresponding refactoring suggestions

124 The rest of this article is organized as follows. Section 2 discusses the related work.
125 Section 3 presents the background of our approach. Section 4 introduces RapidOOD
126 in detail, including how to automatically generate design models from requirements
127 models, how to evaluate design quality, and how to give refactoring suggestions. Section
128 5 presents the evaluation of our approach. Section 6 concludes this article.

129 **2 Related Work**

130 According to the existing research, the automatic generation of software design mod-
131 els can be divided into two types according to the input and generation methods. In
132 some research, the input is mainly the requirement described by natural language.
133 The requirement description is processed using the related natural language processing
134 methods to generate the design model. In other research, the input is the requirement
135 model, which uses the model-driven method to convert requirement models to design
136 models by defining the transformation rules. Each study presents requirements and
137 design models differently but contains consistent information. Requirement models
138 generally describe what entities exist in the system, the relationships between them,
139 what services the system can provide to users, what operations the system can com-
140 plete, and the system’s state before and after operations, which information users can
141 obtain from outside the system. Design models generally describe information related
142 to system implementation, such as how classes within the system are organized, how
143 classes relate to each other, and how system operations are accomplished by instances
144 of classes in collaboration.

145 **2.1 Design Generation from Requirements Description**

146 In terms of generating design models from requirements described in natural language,
147 Mala et al.[11] proposed a method to extract object-oriented elements from natural
148 language. This method divides the original requirements into sentences, labels each
149 word in the sentence, and then recombines nouns and verbs through a combiner based
150 on simple phrase grammar to obtain a relatively structured requirements directory.
151 Then, words with different parts of speech are translated into classes, attributes, and
152 use cases through the defined conversion rules. For example, nouns can be translated
153 into classes and attributes of classes. The disadvantage of this work is that the identi-
154 fied classes may be more than the actual classes. The generated class does not contain
155 methods, cannot describe the interaction between objects within the system, and does
156 not contain the information corresponding to the sequence diagram.

To reduce the ambiguity caused by the rich semantics of the natural language, V. s. sharma[12] and others proposed a method to extract functional designs from relatively structured restricted natural languages and generate UML class diagrams and implemented this method as a tool FDCT (functional design creation tool), which was integrated into IBM Rational Software Architect as a plug-in. The original requirement text is processed by the tool rat produced by its previous work to obtain specific types of strings. The heuristic rules proposed by it are applied to these strings, and finally, a class diagram containing classes, methods and relationships between classes is obtained. Its original requirement text is limited by user-defined vocabulary, and the definition of vocabulary is complex, which is not easy compared with requirement modeling. Only some types of requirements are supported. The generated class diagram does not contain attributes. The internal behavior of the system was not explicitly designed.

Approach *aToucan*[13] automatically generates analysis models consisting of class, sequence, and activity diagrams from requirement models containing natural language. Although *aToucan* can generate sequence diagrams and activity diagrams, their content is limited to human-system interactions and lacks designs to collaborate objects within the system.

2.2 Design Generation from Requirements Model

Work on lightweight formal requirements models provides a foundation for deriving design models. Easterbrook et al.[14] examined the use of lightweight formal methods in requirements modeling through three spacecraft fault-protection case studies. Their work illustrates the value of formalization at the requirements stage, rather than presenting a direct requirements-to-design transformation.

Li et al.[15] proposed a UML-based method for formal requirements modeling. Subsequent work on the Refinement of Component and Object Systems (rCOS)[16–18] provided formal definitions spanning requirements, design, and verification for component-based object-oriented systems. It also described a manual refinement from a requirements model to a detailed functional design model. These studies establish a formal basis for connecting requirements and design, but the refinement described here is not an automated design-generation procedure.

Koch et al.[19] addressed model transformation for web systems. Their input is a WebRE requirements model, and their output is a UWE design model covering presentation, navigation, and content. The mapping between the two metamodels is specified in QVT. This work demonstrates a domain-specific requirements-to-design transformation, while its web-system design focus does not cover the encapsulation of system operations or internal object interactions considered by RapidOOD.

Mussbacher et al.[20] transformed scenario-based aspect-oriented use case maps into UML-based reusable aspect models for reactive systems. This provides another example of requirements-to-design transformation, with input and output models tailored to a different application domain.

Taken together, these studies cover lightweight formal requirements, formal refinement, and domain-specific model transformations. RapidOOD builds on this line of

200 work by targeting both the class structure and the object-interaction sequences of an
201 object-oriented design from its specified requirements-model inputs.

202 2.3 OO Design Quality Metrics and Quality Optimization

203 There are four groups of widely accepted design metrics (CK[21] LK[22], MOOD[23],
204 and QMOOD[24]) for software object-oriented design quality metrics and assessment.
205 The latest QMOOD is an improved hierarchical model for evaluating high-level design
206 quality attributes in object-oriented design. The model relates low-level statistical
207 metrics of design with design properties (such as encapsulation, modularity, coupling,
208 and cohesion) and high-quality properties (such as reusability, flexibility, and com-
209 plexity). Pham[25] et al. used the above four groups of design metrics to propose a
210 set of unified metrics covering many aspects of object-oriented software design. Out-
211 side of the set of quality metrics, there is ongoing exploration into different aspects
212 of object-oriented design. In the research of design quality assessment using design
213 metrics, the existing problem is that there are empirical values when the lower-level
214 quality metrics or features are integrated into the upper-level quality attributes, and
215 these empirical values cannot be automatically adjusted according to the system type
216 and domain knowledge. Meng Yan[26] et al. conducted a systematic mapping study
217 on the related research of quality assessment model, and an important conclusion was
218 that there was a lack of tool support in software design assessment.

219 Regarding software design optimization, there are two paths: quality-driven model
220 transformation and search algorithm. Insfran [27] et al. proposed design guidelines for
221 developing quality-driven model transformations. By using quality models to express
222 quality attributes and transformation models to express alternative model transfor-
223 mations and their effects on quality attributes, transformation designers are allowed
224 to use quality attributes as decision factors for choosing alternative transformations.
225 The correctness of this approach depends on its transformation model and is not flex-
226 ible enough. The use of search algorithms for software design optimization is mainly
227 the use of genetic algorithms [28–30], including single-objective and multi-objective
228 genetic algorithms. Design optimization schemes proposed using model transforma-
229 tion or search algorithms also fail to incorporate system type and domain knowledge
230 and are less effective.

231 2.4 Summary of Related Work

- 232 • Some existing works provide supplementary information about the system’s internal
233 static implementation (including how classes are organized and what operations they
234 have) but fail to include the system’s internal dynamic design information (mainly
235 how objects within the system interact to complete operations).
- 236 • Existing works do not consider the executability of system operation contracts and
237 the sequencing of atomic operations.
- 238 • Existing works do not explain the impact of each iteration on design quality in
239 an easily understandable way, nor do they integrate the quality assessment and
240 refactoring phases in an automated manner.

- To the best of our knowledge, existing work has not effectively combined model-driven methods with LLMs for application in software design tasks.

2.5 Comparison to RM2DM

Our previous tool, RM2PT, facilitated rapid prototyping from requirement models to swiftly confirm requirements. Building on these confirmed requirements, our previous work, RM2DM[31], introduces a tool that can swiftly generate design models adhering to the anemic model design concept from the confirmed requirements. Despite the anemic model design being considered an anti-pattern, it is extensively employed in practical project development, such as in Java EE and Spring, which are both based on the anemic model concept. In comparison to our previous work, the method and tool presented in this paper offer the following:

- *Detailed transformation rules from requirements to sequence diagrams in design models:* Our previous work only provided tool implementation without explicitly stating the rules for generating sequence diagrams in the design models. In this paper, we detail the transformation rules and their use cases.
- *Integration of automatic quality assessment and refactoring methods based on the generated design models:* Our previous work could only generate a simple initial design model. In this method, we use QMOOD metrics and some rules to provide automatic quality assessment and refactoring capabilities, further addressing the issue of design automation. Corresponding tool support is also provided based on Sirius.

Table 1 summarizes the key differences between RM2DM and the approach presented in this paper across five dimensions.

Table 1 Comparison of Key Capabilities between RM2DM and This Work

Dimension	RM2DM	This Work
Class Diagram Generation	✓ Already in RM2DM	✓ Already in RM2DM
Explicit Sequence Diagram Rules	–	New in This Work
Quality Assessment & Refactoring	–	New in This Work
LLM-based Refactoring Suggestions	–	New in This Work
Requirement Traceability	✓ Already in RM2DM	✓ Already in RM2DM

3 Preliminary

3.1 Requirements and Design Models

The requirements model adopted in our approach consists of four components, as illustrated in Figure 1:

- **Use Case Diagram:** Describes interactions between actors and the system to capture user goals.
- **Conceptual Class Diagram:** Depicts domain concepts and their relationships, without specifying operations or implementation details.

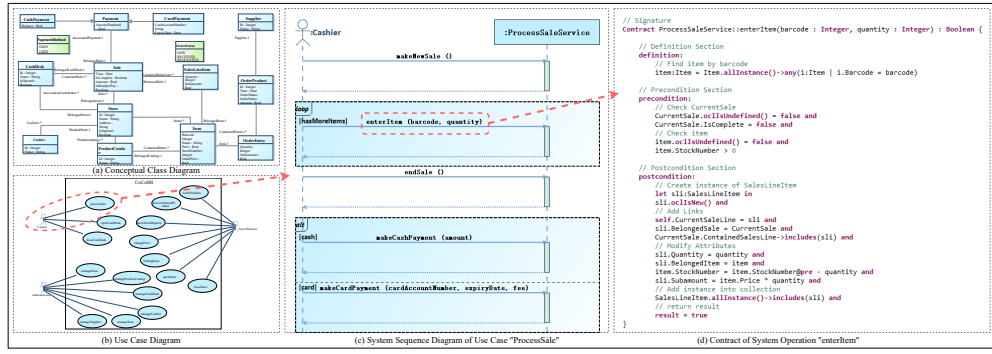


Fig. 1 Requirements Model

- **System Sequence Diagrams:** Shows the sequence of system operations for each use case, treating the system as a black box.
- **Contracts of System Operations:** Formally specifies pre- and post-conditions of each system operation using OCL[3–5, 8, 10].

The design model generated by our approach comprises a class diagram and sequence diagrams. The class diagram defines the static structure, including classes, attributes, operations, and relationships such as aggregation, composition, and inheritance. Sequence diagrams capture the dynamic interactions among objects by detailing the messages exchanged to realize each system operation. The key distinction from the requirements model is that design-level class diagrams specify operations and encapsulation strategies, while sequence diagrams reveal internal object interactions rather than treating the system as a black box. Examples of the generated design model are shown in Figures 2, 3, and 4.

3.2 Anemic Domain Model

The anemic domain model [32] structures the system into three layers: Domain (entities and value objects with data and basic operations), Service (business logic), and Repositories (data persistence and querying). Unlike rich domain models where entities encapsulate both data and business logic, the anemic model places business logic in the Service layer, making entities thin data containers. Despite being considered a design anti-pattern, the anemic model is widely adopted in practice, particularly in web applications where business logic primarily involves CRUD operations, and is supported by mainstream frameworks such as Java EE and Spring. Our approach targets this prevalent architectural style.

3.3 Rule-based Agents and LLMs-based Agents

Software agents and multi-agent systems have received considerable research attention since the 1980s[33]. An agent, as an autonomous system, is capable of interacting with other agents to fulfill its designed objectives. Multi-Agent Systems (MAS) consist of multiple autonomous entities, or agents, that interact within a shared environment

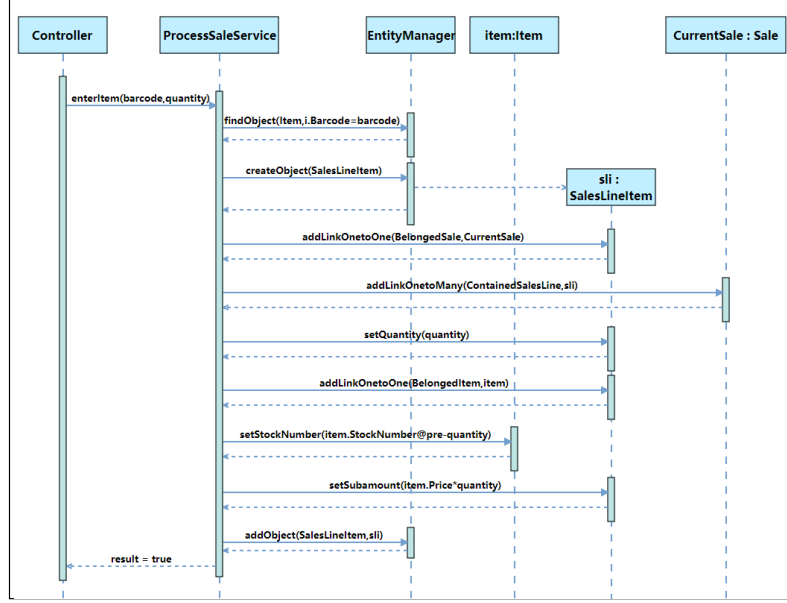


Fig. 4 Generated Sequence Diagram for System Operation “enterItem”

to achieve individual or collective goals. In the research of agent-oriented software engineering, the primary focus is on how to utilize agents as fundamental building modules for designing and developing complex systems, with an emphasis on modularity, reusability, and software architecture. The agents therein typically operate based on predefined rules or logical reasoning, and their interactions mainly base on rule and deterministic protocols or algorithms. We refer to this kind of agent as Rule-based Agents.

Large Language Models are a class of deep learning models designed to understand and generate human-like text. LLMs leverage massive datasets and billions of parameters to achieve remarkable language comprehension and generation capabilities. Due to the rapid development of LLMs, LLMs-based agents have attracted significant attention. LLMs-based agents take LLMs as the core. Through reinforcement learning and function-calling techniques, such agents can automate complex tasks via the “perceive-decide-act” process.

4 Approach

This section describes our method for transforming requirement models into design models (including a class diagram and sequence diagrams), assessing design quality during the design phase, and offering straightforward refactoring recommendations.

4.1 Overview

Our approach takes validated requirements models as input. Auto-prototyping work, such as RM2PT, assists in swiftly attaining a validated requirements model by

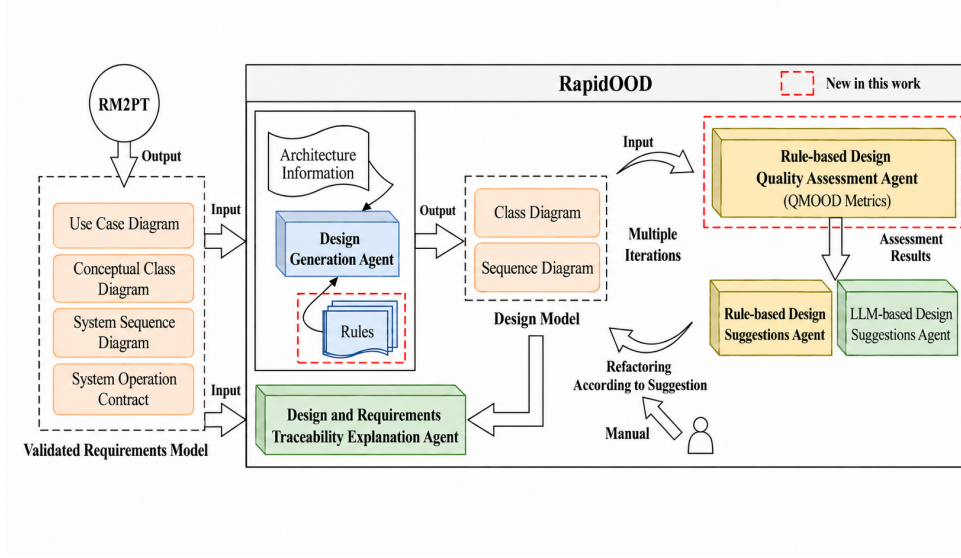


Fig. 5 Overview of RapidOOD. Red dashed boxes indicate capabilities newly introduced in this work compared with RM2DM.

321 automatically producing prototypes from requirement models, thereby facilitating
 322 incremental and agile requirements validation. We propose a set of rules and algo-
 323 rithms to generate the design models composed of a class diagram and a set of sequence
 324 diagrams. Based on the initial design models generated, we employ QMOOD, a hierar-
 325 chical model for evaluating the quality of object-oriented designs, to assess the design
 326 quality. The results of the quality assessment metrics and the simple refactoring sug-
 327 gestions we propose are fed back to the users. Users can then manually iterate on the
 328 design based on this feedback, thereby swiftly achieving a relatively higher quality
 329 design.

330 In this paper, the agents are not solely centered around LLMs, instead, the frame-
 331 work incorporates a hybrid approach where agents are built upon both LLM-driven
 332 capabilities and traditional model-driven methods. While LLM-based agents excel
 333 in tasks requiring natural language understanding, reasoning, and adaptability, the
 334 model-driven agents leverage predefined rules, domain-specific models, and structured
 335 decision-making processes to handle tasks with deterministic logic or stringent perfor-
 336 mance requirements. This duality allows the system to balance the flexibility of LLMs
 337 with the reliability and efficiency of traditional approaches, creating a more robust
 338 and versatile multi-agent framework.

339 In RapidOOD, we designed the Design Generation Agent, Design Quality Assess-
 340 ment Agent, and rule-based Design Suggestions Agent that provides simple recom-
 341 mendations using a model-driven, rule-based approach. The model-driven approach
 342 generates the initial design model from the requirements model, ensuring excellent
 343 traceability and the scale of the initial design. (As can be seen in the subsequent
 344 experimental section, the design models generated solely by LLMs are limited in scale

345 and often lack essential relationships or operations.) For the quality assessment of the
346 design, a quantitative method based on QMOOD Metrics was adopted. By statistically
347 analyzing the model elements of the design model, high-level quality attribute met-
348 rics of the design model were obtained, such as reusability. For low quality attributes,
349 rule-based design refactoring suggestions were provided. For example, parts with high
350 coupling in the design model were highlighted and suggestions for refactoring were
351 given.

352 We designed the LLM-Based Design Suggestions Agent and the Traceability Expla-
353 nation Agent. The aim is to utilize the knowledge-emergence ability of LLMs to
354 provide expert-level design refactoring suggestions and to offer easily understandable
355 explanations for the traceability between the design and the requirements.

356 4.2 Design Model Generation Agent

357 4.2.1 Transformation Rules of Class Diagram Generation

358 Three transformation rules are proposed to automatically generate a class diagram
359 from the requirement model .

360 *Preserve the Classes and Relationships*

361 The first rule is to preserve the relationships between classes and classes in the concep-
362 tual class diagram and add GET/SET methods corresponding to the class attributes.
363 This design is based on the fact that the relationships between classes and classes in
364 the conceptual class diagram in the requirement model are naturally consistent with
365 the actual situation, and this part should remain the same; otherwise, relationship
366 errors or missing relationships will occur.

367 *Generate Service Class*

368 The second rule is to encapsulate all use case-related system operations into a new Ser-
369 vice class based on the use case diagram (including use cases) and the system sequence
370 diagram (including system operations related to use cases). For a case, according to
371 the information in the use case diagram and the system sequence diagram, a new class
372 is extracted, which contains the class name, attributes, operations, and other infor-
373 mation. It extracts class names from use case diagrams and attributes and operations
374 from system sequence diagrams and system operation contracts.

375 The classes extracted by this rule will serve as the content of the Service layer and
376 can be used to provide services to the outside world. Service will act as the initiator
377 of the system operation, and then the various classes within the system will work
378 together to complete the various system operations.

379 *Generate EntityManager Class*

380 The third rule involves the creation of a new EntityManager class, which serves as the
381 central hub for all other classes. It is responsible for adding, deleting, modifying, and
382 querying objects, and thus should have primitive operations related to these tasks.
383 The operations owned by EntityManager are derived by summarizing the OCL state-
384 ments in the system operation contract. 10 types of OCL expressions associated with

385 EntityManager and 7 primitive operations are needed.

$$\begin{aligned} E_1 : obs : Set(ClassName) &= ClassName.allInstances() \\ E_2 : obs : Set(ClassName) &= ClassName.allInstances() \\ &- > select(o|conditions(o)) \end{aligned}$$

386 The meaning of statement 1 is to find instances of all ClassName classes in the
387 system, and the meaning of statement 2 is to find instances of conditional conditions
388 of all ClassName classes in the system. To complete the behavior described by the two
389 OCL statements, the class EntityManager requires the operation findObjects().

$$\begin{aligned} E_3 : ob : ClassName &= ClassName.allInstances() \\ &- > any(o|conditions(o)) \end{aligned}$$

390 The meaning of statement 3 is to find an instance of a conditional condition of
391 any ClassName class in the system. Therefore, the class EntityManager requires a
392 findObject() operation.

$$\begin{aligned} E_4 : o : ClassName &= ob.assoName \\ E_5 : o : ClassName &= ob.assoName \\ &- > any(o|conditions(o)) \end{aligned}$$

393 Statement 4 means to find a single instance of a class in the system that has an
394 assoName relationship with an ob class, and statement 5 means to find any instance
395 of a conditional condition of a class that has an assoName relationship with an OB
396 class in the system. Both statements find a single object based on the association
397 relationship, so the EntityManager needs an operation findLinkedObject ().

$$\begin{aligned} E_6 : obs : Set(ClassName) &= ob.assoName \\ E_7 : obs : Set(ClassName) &= ob.assoName \\ &- > select(o|conditions(o)) \end{aligned}$$

398 The meaning of statement 6 is to find a collection of instances of a class in the
399 system that has an assoName relationship with an ob class, and the meaning of state-
400 ment 7 is to find a collection of instances of a conditional class in the system that has
401 an assoName relationship with an ob class, so the EntityManager needs to be able to
402 find a collection of objects based on the association relationship and needs to operate
403 findLinkedObjects ().

$$E_8 : let ob : ClassName in ob.oclIsNew()$$

404 The meaning of statement 8 is to define an object OB for a ClassName, so the
405 EntityManager needs to be able to create instances of a particular class and needs to

406 have an operation `createObject ()`.

$$E_9 : \text{ClassName.allInstances()} - > \text{includes}(ob)$$

407 Statement 9 appears in the Postcondition section of the system operation contract,
 408 indicating that the OB object needs to be guaranteed to be included in the instance of
 409 ClassName after the system operation is executed, so EntityManager needs to add the
 410 OB object to the list of instances of ClassName class, and the `addObject ()` operation
 411 is required.

$$E_{10} : \text{ClassName.allInstances()} - > \text{excludes}(ob)$$

412 Statement 10 appears in the Postcondition section of the system operation contract,
 413 indicating that you need to ensure that the ClassName instance after the system
 414 operation is executed does not contain an ob object, so EntityManager needs to
 415 remove the OB object from the list of ClassName instances, and you need an operation
 416 `releaseObject()`.

417 4.2.2 Transformation Rules of Sequence Diagram Generation

418 RapidOOD uses sequence diagrams to describe the collaboration of objects within the
 419 system. The main content of the sequence diagram is the messages passed between
 420 objects. Based on the anemic domain model, there is a controller in the system. The
 421 controller sends the invocation messages of system operations to the corresponding
 422 service class. The service class sends messages to the related objects inside the system,
 423 and the objects collaborate to complete the system operations and return. The primary
 424 challenge lies in the generation of messages between objects. Of the three components
 425 of the system operation contract, the OCL expressions in the precondition part define
 426 the system's required state before executing the operation. As these preconditions do
 427 not involve interactions between class instances, they are not converted into messages.
 428 The definition and post-condition of the system operation contract, on the other hand,
 429 describe the system's dynamic behavior. The OCL expressions in these sections can be
 430 transformed into messages in sequence diagrams. To facilitate this transformation, a
 431 set of 17 transformation rules has been established. Transformation rules are presented
 432 in this form:

$$\text{Rule} : \frac{\text{OCL Expression}}{\text{Message Label, Sender, Receiver}}$$

433 The transformation rule contains two parts: the above is an OCL expression in
 434 the contracts, and the bottom part is the generated message and the message's sender
 435 and receiver.

$$R1 : \frac{\text{obs: Set (ClassName) = ClassName.allInstances()}}{\text{findObjects (ClassName), Service, EntityManager}}$$

$$R2 : \frac{\text{obs: Set (ClassName) = ClassName.allInstances()} \rightarrow \text{select (o | conditions(o))}}{\text{findObjects (ClassName, conditions (o)), Service, EntityManager}}$$

$$R3: \frac{ob: \text{ClassName} = \text{ClassName.allInstances}() \rightarrow \text{any } (o \mid \text{conditions}(o))}{\text{findObject } (\text{ClassName}, \text{conditions } (o)), \text{Service}, \text{EntityManager}}$$

$$R4: \frac{o: \text{ClassName} = ob.assoName}{\text{findLinkedObject } (ob, assoName), \text{Service}, \text{EntityManager}}$$

$$R5: \frac{o: \text{ClassName} = ob.assoName - \dot{\exists} \text{ any } (o - \text{conditions}(o))}{\text{findLinkedObject } (ob, assoName, \text{conditions}(o)), \text{Service}, \text{EntityManager}}$$

$$R6: \frac{obs: \text{Set } (\text{ClassName}) = ob.assoName}{\text{findLinkedObjects } (ob, assoName), \text{Service}, \text{EntityManager}}$$

$$R7: \frac{obs: \text{Set } (\text{ClassName}) = ob.assoName - \dot{\exists} \text{ select } (o - \text{conditions}(o))}{\text{findLinkedObjects } (ob, assoName, \text{conditions}(o)), \text{Service}, \text{EntityManager}}$$

Rules 1 to 7 are the conversion rules from the definition part of the OCL statement to the message. The meaning of this part of the statement has been introduced above. According to this part of the conversion rules, the sending end of the message is Service, and the receiving end is EntityManager.

$$R8: \frac{\text{let } ob: \text{ClassName} \text{ in } ob.\text{oclIsNew}()}{\text{createObject } (\text{ClassName}), \text{EntityManager}, ob}$$

The meaning of statement 8 is to generate an instance ob of the ClassName class. The operation of creating an object is initiated by the Service and completed by the EntityManager to generate an object ob. There are actually two request messages in this process. One is that the Service sends a message to the EntityManager to request the creation of an instance; the other is the creation message of the object ob created by the EntityManager.

$$R9: \frac{\text{ClassName.allInstances}() - \dot{\exists} \text{ includes } (ob)}{\text{addObject } (\text{ClassName}, ob), \text{Service}, \text{EntityManager}}$$

$$R10: \frac{\text{ClassName.allInstances}() - \dot{\exists} \text{ excludes } (ob)}{\text{releaseObject } (\text{ClassName}, ob), \text{Service}, \text{EntityManager}}$$

Statement 9 and statement 10 are the OCL statements of the postcondition part, and the related class is also EntityManager, which has been introduced in the previous article. For the messages obtained according to rules 9 and 10, the sending end is Service, and the receiving end is EntityManager.

$$R11: \frac{ob.assoName - \dot{\exists} \text{ includes } (addOb)}{\text{addLinkOnetoMany } (assoName, addOb), \text{Service}, ob}$$

The meaning of statement 11 is to add the addOb object to the list of objects pointed to by the assoName relation of ob, corresponding to the atomic operation addLinkOnetoMany(), which is executed by the object ob.

$$R12: \frac{ob.assoName - \dot{\exists} \text{ excludes } (removeOb)}{\text{removeLinkOnetoMany } (assoName, removeOb), \text{Service}, \text{EntityManager}}$$

The meaning of statement 12 is to remove the removeOb object from the list of objects pointed to by the assoName relation of ob, and the corresponding atomic operation is removeLinkOnetoOne(), which is executed by the object ob.

$$R13: \frac{ob.assoName = addOb}{\text{addLinkOnetoOne } (assoName, addOb), \text{Service}, ob}$$

456 The meaning of statement 13 is to change the object pointed to by the assoName
 457 relation of ob to addOb, and the corresponding atomic operation is addLinkOne-
 458 toOne(), which is executed by the object ob.

$$R14 : \frac{ob.assoName = null}{removeLinkOnetoOne(assoName), Service, ob}$$

459 The meaning of statement 14 is to change the object pointed to by the assoName
 460 relation of ob to null, i.e., it does not point to any object, and the corresponding
 461 atomic operation is removeLinkOnetoOne(), which is executed by the object ob.

$$R15 : \frac{ob.attriName = mathExp}{setAttriName(mathExp), Service, ob}$$

462 The meaning of statement 15 is to change the value of the attriName attribute of
 463 ob to the value represented by mathExp, and the corresponding atomic operation is
 464 setAttribute(), which is executed by the object ob.

$$R16 : \frac{result = var}{return(var), Service, Controller}$$

465 The meaning of statement 16 is to return the result of the system operation.

$$R17 : \frac{ThirdPartyService.opName(vars)}{opName(vars), Service, ThirdPartyService}$$

466 The meaning of statement 17 is to call a method of a third-party service. For each
 467 OCL expression in the system operation contract, apply rules 1-7 if it belongs to the
 468 definition part of the contract and rules 8-17 if it belongs to the post-condition part.
 469 Determine the rules for the OCL expression based on its syntactic structure. Apply
 470 the rules to extract the label, the sending object and receiving object of the message
 471 from the OCL expression.

Algorithm 1 Sequence Diagram Messages Generation

Input: System Operation Contract

Output: Sequence Diagram Messages

```

1: for each OCLExpression ∈ Contract do
2:   if OCLExpression ∈ definition then
3:     applyRule(1 – 7)
4:   else if OCLExpression ∈ postcondition then
5:     applyRule(8 – 17)
6:   end if
7: end for
8: procedure applyRule(ruleOrder)
9:   setMessageLabel()
10:  setMessageSender()
11:  setMessageReceiver()

```

472 Algorithm 1 shows how to apply the rules to generate messages from system opera-
 473 tion contracts. For each OCL expression in the system operation contract, apply rules
 474 1-7 if it belongs to the definition part of the contract and rules 8-17 if it belongs to

475 the post-condition part. Determine the rules for the OCL expression based on its syn-
476 tactic structure. Apply the rules to extract the label, the sending object and receiving
477 object of the message from the OCL expression.

478 Algorithm 2 describes how to order OCL statements to ensure that they are
479 executable. It involves constructing a dependency graph and performing topological
480 sorting to ensure that all dependencies are respected. The algorithm details are as
481 follows.

482 **Initialize the Dependency Graph:** Create an empty directed graph $G = (V, E)$,
483 where V represents the set of nodes (OCL statements) and E represents the set of
484 directed edges (dependencies between statements).

485 **Add Nodes to the Graph:** For each OCL statement $s_i \in S$ (where S is the set
486 of OCL statements), add s_i as a node in V .

487 **Add Edges to the Graph:** For each OCL statement $s_i \in S$, identify its depen-
488 dencies. For each dependency s_j of s_i , add a directed edge (s_j, s_i) to E . This indicates
489 that s_i depends on s_j .

490 **Topological Sorting Initialization:** Initialize an empty list L to store the sorted
491 OCL statements. Initialize an empty set S to store all nodes with no incoming edges.

492 **Identify Nodes with No Incoming Edges:** For each node $v \in V$, if v has no
493 incoming edges, add v to S .

494 **Topological Sorting Process:** While S is not empty: - Remove a node n from
495 S . - Add n to L . - For each node m such that $(n, m) \in E$: - Remove edge (n, m) from
496 E . - If m has no other incoming edges, add m to S .

497 **Cycle Detection:** If there are any remaining edges in E , this indicates that the
498 graph G contains at least one cycle, and a topological sorting is not possible. In this
499 case, report an error.

500 **Return the Sorted List:** Return the list L as the topologically sorted order of
501 OCL statements.

Algorithm 2 OCL Statement Execution Order Sorting

Input: A set of OCL statements $S = \{s_1, s_2, \dots, s_n\}$ with dependencies

Output: A topologically sorted list of OCL statements

```
1: Initialize an empty dependency graph  $G = (V, E)$ 
2: for each  $s_i \in S$  do
3:   Add  $s_i$  as a node in  $V$ 
4: end for
5: for each  $s_i \in S$  do
6:   for each  $s_j \in \text{dependencies}(s_i)$  do
7:     Add a directed edge  $(s_j, s_i)$  to  $E$ 
8:   end for
9: end for
10: Initialize an empty list  $L$  to store the sorted OCL statements
11: Initialize an empty queue  $Q$  to store all nodes with no incoming edges
12: for each  $v \in V$  do
13:   if  $v$  has no incoming edges then
14:     Add  $v$  to  $Q$ 
15:   end if
16: end for
17: while  $Q$  is not empty do
18:   Remove a node  $n$  from  $Q$ 
19:   Add  $n$  to  $L$ 
20:   for each node  $m$  such that  $(n, m) \in E$  do
21:     Remove edge  $(n, m)$  from  $E$ 
22:     if  $m$  has no other incoming edges then
23:       Add  $m$  to  $Q$ 
24:     end if
25:   end for
26: end while
27: if any edges remain in  $E$  then
28:   error: Graph  $G$  has at least one cycle
29: end if
30: return  $L$ 
```

4.3 Quality Assessment and Refactoring Recommendations of Design Model

We evaluated design quality using QMOOD, a hierarchical model to evaluate object-oriented design quality. In the QMOOD method, the high-level quality attribute indicators are calculated from the low-level element statistical indicators. Traditionally, this quality assessment is based on code. However, we noticed that many of these underlying metrics were available early in the design phase of the project, empowering designers to make proactive decisions. For example, QMOOD uses the number of

510 classes to represent the size of the design, which is determined when the design is com-
511 pleted. Another example is the number of levels in a hierarchical architecture, whose
512 value is fixed and can also be obtained at design time.

513 4.3.1 Rule-based Design refactoring Agent

514 Based on the results of the quality attributes assessment, we provide simple rule-based
515 design refactoring recommendations to designers. Since the calculation of high-level
516 quality attribute indicators is based on the statistical number of underlying elements,
517 we can identify the design elements that most impact a particular quality attribute
518 indicator. We decided to set the granularity of design elements to classes, and when
519 a class was found to have the most significant negative impact on specific quality
520 attribute metrics, we highlighted it and suggested that designers make refactorings.

- 521 • Design size: Consider the case where the number of classes exceeds a threshold that
522 indicates complexity. This metric is available early in the design phase. If the number
523 of classes is too high, we recommend strategies such as refactoring into smaller
524 modules or applying design patterns like the Factory pattern to manage complexity.
- 525 • Hierarchy: In a hierarchical architecture, if the number of levels is too deep, indi-
526 cating a potential for maintenance issues, we suggest flattening the structure by
527 merging similar classes or using composition over inheritance where appropriate.
- 528 • Coupling: When the coupling between classes is too high, indicating a lack of mod-
529 ularity, we recommend introducing interfaces or applying the Dependency Injection
530 pattern to decouple components.
- 531 • Cohesion: If the cohesion within a class is low, suggesting that the class has multiple
532 responsibilities, we advise splitting the class into smaller, more focused classes or
533 applying the Single Responsibility Principle.
- 534 • Polymorphism: If there is a high degree of polymorphism, indicating a com-
535 plex inheritance hierarchy, we recommend simplifying the inheritance structure by
536 reducing the depth of the hierarchy or employing composition instead of inheritance.

537 For example, in the generated class diagram shown in Figure 2, we can see that the
538 EntityManager class has the most significant number of directly related classes, which
539 has the highest degree of coupling according to the metric calculation, thus suggest-
540 ing a design refactoring for it: “EntityManager has high Direct Class Coupling, please
541 pay attention to proper class responsibility division”. Considering the refactoring of
542 EntityManager, in this design, EntityManager is mainly responsible for receiving the
543 requests of the service layer and modifying the attributes and association relations of
544 the related classes in the entity layer. The EntityManager can be split into multiple
545 repositories, each responsible for modifying a class. This reduces the degree of coupling
546 between classes, and the responsibility of each class is more single, in line with the
547 single responsibility principle. We can see that this modification has changed some of
548 the low-level design metrics and affected the high-level design quality. Understandabil-
549 ity decreases because multiple managers are more complex than a single Manager in
550 terms of design structure. However, splitting it into various managers makes for better
551 readability. Functionality goes up because multiple Managers bring access to multiple

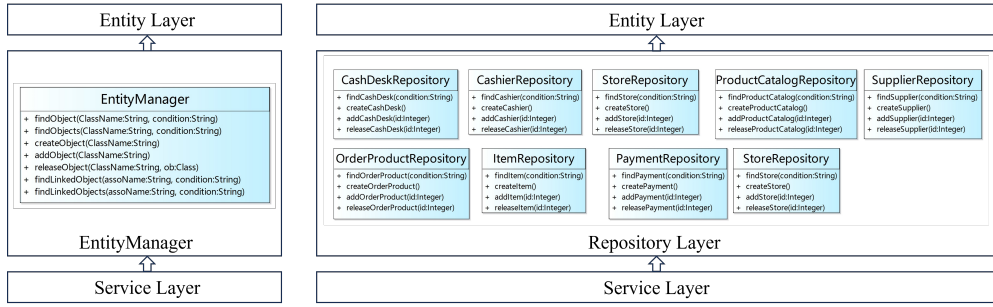
Table 2 Comparison of Design Quality Indicators Before and After Modification

Metric	DSC	NOH	ANA	DAM	DCC	CAM	MOA	MFA	NOP	CIS	NOM
Before	24.0000	3.0000	0.0833	1.0000	2.7917	1.0000	5.0000	0.4500	1.0000	144.0000	144.0000
After	32.0000	3.0000	0.0625	1.0000	2.0625	1.0000	5.0000	0.4500	1.0000	173.0000	173.0000
After/Before	1.3333	1.0000	0.7503	1.0000	0.7388	1.0000	1.0000	1.0000	1.0000	1.2014	1.2014

Table 3 Comparison of Design Quality Before and After Modification

Metric	Resuability	Flexibility	Understandability	Functionality	Extendibility	Effectiveness
Before	1.0000	1.0000	-0.9900	1.0000	1.0000	1.0000
After	1.3327	1.0653	-0.9979	1.1176	1.0058	0.9501

common interfaces. Less coupling between classes also leads to better reusability, flexibility, and extensibility. This example illustrates the effectiveness of the refactoring method according to quality metrics.

**Fig. 6** Optimized Class Diagram

4.3.2 LLM-based Design refactoring Agent

The rapid development of Large Language Models (LLMs), such as GPT, BERT, and their successors, has revolutionized various domains by providing unprecedented capabilities in natural language understanding and generation. These models are not only reshaping industries like customer support, content creation, and data analytics but are also increasingly being leveraged in specialized technical domains. In the realm of Object-Oriented (OO) design refactoring, LLMs have demonstrated their potential to analyze complex software models, identify design inefficiencies, and suggest improvements grounded in best practices and design principles such as SOLID and GRASP. By integrating domain-specific knowledge with pattern recognition, LLMs provide actionable recommendations, enhance maintainability, and reduce technical debt, making them a valuable tool for modern software engineering. In our approach, GPT-based capabilities are utilized in three key aspects:

- Providing design refactoring recommendations based on the current quality metrics.

Model refactoring Suggestion

PROMPT

The plantUML of Design is as follows: [DesignUML]. The design quality of this design are as follows: [DesignQuality].

Based on design patterns (e.g., Singleton, Factory, Observer, etc.) and GRASP design principles (e.g., Information Expert, Low Coupling, High Cohesion), suggest improvements for the model. Please explain the following:

- 1. Which classes or relationships can be optimized? Provide specific suggestions for changes.
- 2. Based on design patterns, which patterns could enhance the maintainability, scalability, or performance of the model?
- 3. How can GRASP principles optimize the assignment of responsibilities and interactions between classes?
- 4. How will the proposed changes address potential issues in the existing model (e.g., high coupling, low cohesion)?
- 5. After modifications, which aspects of the model (e.g., class responsibilities, relationships, coupling) will improve?

Please analyze each suggestion step by step (Chain of Thought), providing an explanation for each and proposing the most suitable solution.

The overall structure should be that the first part is a series of specific change suggestions, and the second part is explanations of the change suggestions.

VARIABLE

[DesignUML]: The plantUML representation of Design Model.

[DesignQuality]: Design quality assessment results automatically computed by the modeling tool, employing the QMOOD metrics, including reusability, flexibility, understandability, functionality, extendibility and effectiveness.

569

570 The provided prompt is designed to facilitate comprehensive and structured analysis
571 of software design models, leveraging the capabilities of Large Language Models
572 (LLMs) to optimize Object-Oriented (OO) design. It integrates principles from
573 design patterns and GRASP (General Responsibility Assignment Software Patterns)
574 to ensure that recommendations align with established best practices in software
575 engineering. By using a PlantUML representation of the design model as input,
576 the prompt focuses on actionable improvements, making it suitable for iterative
577 development processes.

578 One of the key strengths of this prompt lies in its focus on clarity and step-by-
579 step reasoning. The Chain of Thought (CoT) methodology is explicitly embedded,
580 encouraging the model to break down complex design quality assessments into man-
581 ageable components. This approach not only improves the interpretability of the
582 recommendations but also ensures that the reasoning behind each suggestion is
583 transparent and traceable. Furthermore, the prompt emphasizes a dual-structured

584 response, first presenting concrete recommendations and then providing detailed
585 justifications. This structure fosters a deeper understanding of how the proposed
586 changes will improve design quality metrics, such as maintainability, scalability, and
587 performance.

588 Another notable advantage is the explicit integration of both design patterns and
589 GRASP principles. By asking the model to evaluate the applicability of patterns
590 such as Singleton, Factory, and Observer, the prompt ensures that the recommenda-
591 tions are grounded in established strategies for solving common design challenges.
592 Simultaneously, GRASP principles guide the analysis of responsibility allocation
593 and interactions between classes, addressing issues like high coupling and low cohe-
594 sion. This dual focus enables a holistic quality assessment of the design, balancing
595 the structural and behavioral aspects of the model.

- 596 • Automatically generating explanations for changes in design quality metrics follow-
597 ing specific operations.

Model refactoring Explanation

PROMPT

The plantUML of Design is as follows: [DesignUML]. Based on the modified UML Class Diagram, explain why these changes will impact the quality of the model. Specifically, explain how the following aspects are improved or degraded:

- 1. Has the distribution of responsibilities among classes become more reasonable? Has cohesion been improved?
- 2. Has the coupling between classes been effectively reduced after the modifications?
- 3. How do the newly added design patterns or GRASP principles affect the scalability, maintainability, or performance of the model?
- 4. Has the design reduced complexity and improved the understandability of the model?
- 5. Overall, have the quality metrics of the model (e.g., flexibility, reusability) been improved? If so, in which specific areas?

Please explain step by step (Chain of Thought) the specific impact of each change.

VARIABLE

[DesignUML]: The plantUML representation of Design Model.

598

599 The accompanying prompt is crafted to enable detailed, step-by-step explanations
600 of the impacts of design modifications on software quality metrics. It provides
601 a structured framework that leverages the analytical capabilities of Large Lan-
602 guage Models (LLMs) to assess how specific changes in a UML Class Diagram
603 influence the overall design quality. By embedding Chain of Thought (CoT) rea-
604 soning, the prompt ensures that each modification is analyzed comprehensively and

605 methodically, making it a powerful tool for understanding and refining software
606 models.

607 A core strength of the prompt lies in its focus on critical aspects of design evaluation,
608 such as cohesion, coupling, scalability, and maintainability. These are foundational
609 metrics in assessing software quality, and the prompt explicitly calls for a systematic
610 analysis of how each metric is impacted by the proposed changes. For example, by
611 addressing whether responsibility distribution has become more logical or whether
612 cohesion has improved, the prompt ensures that the analysis is grounded in prin-
613 ciples of Object-Oriented Design (OOD). Similarly, the explicit focus on coupling
614 reduction and the application of design patterns or GRASP principles highlights
615 the importance of achieving balance in structural and behavioral design.

616 Moreover, the prompt emphasizes actionable insights by asking how design pat-
617 terns and GRASP principles enhance the model’s scalability, maintainability, and
618 performance. This ensures that the explanations are not only theoretical but also
619 practical, providing developers with a clear understanding of the trade-offs and ben-
620 efits introduced by specific design decisions. The inclusion of considerations like
621 complexity reduction and understandability further aligns the prompt with real-
622 world development goals, ensuring that the resulting models are both robust and
623 developer-friendly.

624 • Analyzing the traceability between the current design and specific requirements,
625 identifying unmet requirements, and suggesting complementary design elements to
626 address the gaps. The prompt is expertly designed to facilitate a comprehensive
627 and structured analysis of traceability between design and requirements in soft-
628 ware engineering. It leverages the capabilities of Large Language Models (LLMs) to
629 ensure precise mapping of requirements to design elements while identifying gaps
630 and proposing actionable solutions. The use of a PlantUML-based representation
631 for the design and a detailed requirements file ensures that the analysis is grounded
632 in clear, visual, and textual artifacts, enhancing the clarity and accuracy of the
633 recommendations.

634 A particularly valuable aspect of the prompt is its focus on unmet requirements.
635 By explicitly asking the model to identify and explain why certain requirements
636 are missing or overlooked, it fosters a deeper understanding of potential gaps in
637 the design process. This ensures that the analysis not only addresses technical
638 shortcomings but also considers contextual factors such as oversight, complexity, or
639 misalignment of goals.

640 The recommendation for complementary design elements further enhances the
641 prompt’s utility. By requiring the model to propose solutions aligned with estab-
642 lished best practices like SOLID principles, the prompt ensures that the suggestions
643 are both practical and theoretically sound. This dual focus on maintainability and
644 scalability reflects a strong alignment with modern software engineering priorities,
645 making the output highly actionable for practitioners.

646 The final step, justification and validation, adds an extra layer of rigor to the anal-
647 ysis. By asking the model to explain the reasoning behind proposed solutions and
648 discuss potential trade-offs, the prompt ensures that the recommendations are bal-
649 anced and well-considered. This not only aids in decision-making but also provides

650 valuable insights into the implications of the proposed changes, such as their impact
651 on design quality metrics like cohesion, coupling, and reusability.

Traceability Between Design and Requirements

PROMPT

The plantUML of Design is as follows: [DesignUML]. The Requirements is as follows: [Remodel].

I am designing a software system and need your expertise in analyzing the traceability between the current design and specific requirements. Please follow these steps to provide a detailed and actionable response:

- 1. Traceability Analysis: Begin by identifying the key requirements from the provided list. Map each requirement to the corresponding design elements.
- 2. Unmet Requirements Identification: Highlight any requirements that are not addressed by the current design. For each unmet requirement, explain why it might be missing or overlooked.
- 3. Recommendations for Complementary Design Elements: For each unmet requirement, propose specific design elements or modifications to the existing design. Ensure that the recommendations align with best practices in software design (e.g., SOLID principles, maintainability, scalability).
- 4. Justification and Validation: Provide reasoning for the proposed design elements, explaining how they fulfill the unmet requirements. Discuss any potential trade-offs or impacts on the overall design quality metrics.

Use a structured format for your response, such as:

Requirement X: (State the unmet requirement)

Current Coverage: (Briefly describe if it's partially covered or missing entirely)

Proposed Solution: (Detail the complementary design element)

Justification: (Explain why this solution is appropriate and its potential effects)

Finally, ensure that your response is concise yet comprehensive, avoiding unnecessary jargon. If additional clarification or details about the current design are needed, specify the information required.

VARIABLE

[plantUML]: The plantUML representation of Design Model.

[Remodel]: The Remodel file of Requirements.

652

653 **Discussion: Why OCL/Java Rather Than ATL or Epsilon.** A natural
654 question is whether a dedicated model transformation language such as ATL or
655 Epsilon would be a more appropriate vehicle for the transformation rules described
656 in this section. While ATL and Epsilon excel at expressing declarative, pattern-
657 matching mappings between metamodels, the rules in our approach involve two types

of logic that are more naturally expressed procedurally. First, the dependency analysis required to determine a valid execution order for OCL statements (Algorithm 2) constructs and traverses a dependency graph; this imperative, graph-centric computation does not map cleanly onto the declarative match-apply paradigm of ATL rules. Second, the sequence diagram generation rules must dynamically instantiate and link objects whose number and types are determined at runtime by the structure of each individual OCL contract, rather than by a fixed metamodel pattern. For these reasons, we implement the transformation logic in Java (integrated with the Sirius modeling workbench), using OCL as a *formal specification layer* to parse and interpret contract semantics. This hybrid strategy retains the readability and formal grounding of OCL for contract analysis while leveraging the full expressiveness of Java for the procedural coordination logic. We acknowledge that this places an additional burden on end users who must author OCL contracts; tooling support for contract authoring and validation (already partially provided by the RM2PT framework) is an avenue for future improvement.

5 Evaluation

In this section, We use six requirement models of different software systems to perform evaluation.

5.1 Research Questions

- **RQ1:** To what extent do the transformation rules proposed by our approach cover various situations?
- **RQ2:** How is the quality of the generated design models, and how does it compare to manually created models?
- **RQ3:** Does our automated approach for constructing design models offer a time efficiency advantage over manual approaches?.

5.2 Experimental Setting

We developed a tool based on the **Eclipse Modeling framework** to automate our approach. We use ATL transformation language to implement the requirements model to design model transformation and Sirius to graphically display and modify the design model. The tool provides the functionality to edit the generated design models. During the design model generation and modification process, automatic metrics calculation and identification of design elements to be optimized are achieved by writing Java Services for Sirius.

To address RQ1, we start by counting the relevant measures within both the requirement model and the automatically generated design model. These counts are then used to calculate specific metrics that are crucial for our analysis.

For RQ2, we enlist the expertise of seasoned professionals in the software engineering field. These experts are carefully recruited based on their extensive experience and in-depth knowledge of design models and domain modeling. They are tasked with establishing design models that conform to an anaemic domain model, derived directly

from the requirement models. These expert-generated models serve as the reference models for our study. To facilitate comparison, we also involve postgraduate students who are specializing in software engineering. These students are recruited through academic channels, focusing on those who have demonstrated proficiency in software design and development in their coursework.

Before the postgraduates begin constructing the design models, they undergo a comprehensive training program. This program is structured to ensure they have a thorough understanding of the tools and methodologies they will be using, as well as the concepts of requirement modeling and anaemic domain models. The training involves hands-on sessions where the participants are introduced to the tools necessary for constructing design models. They also receive detailed explanations of the models they are expected to build, focusing particularly on the principles of the anaemic domain model and how it aligns with the requirement model.

Once the training is complete, the participants are provided with a specific set of requirement documents. Their task is to construct a design model based on these documents, adhering to the principles of the anaemic domain model. The design model they are asked to create is for a monolithic system built using SpringBoot. This part of the experiment is critical as it allows us to directly compare the manually constructed design models with those generated automatically, thereby providing insights into the effectiveness and efficiency of each approach.

For RQ3, we focus on the time efficiency of the design model construction process. We meticulously record the time taken by the postgraduates to manually construct the design model. This duration is then compared with the time required to build the model using our automated method. The comparison of these time metrics provides valuable data on the efficiency gains that our automated approach offers over traditional manual methods.

5.3 Case Studies

The dataset has six requirement models, four of which come from RM2PT. The cases of the six requirement models describe the requirements of six common scenarios, including automatic teller machine(ATM), supermarket system(CoCoME), library management system(LibMS), loan management system(LoanPS), Parking spot share system(ParkShare) and Airport management system(AirMS). The case of an automatic teller machine mainly involves the user's need to use an ATM to deposit and withdraw money. It is relatively simple and can be used as a rapid preliminary verification of the automatic generation method of the design model. The case of CoCoME describes the needs of the cashier in the supermarket to process the shopping documents and the administrator to manage the supermarket inventory. The process involved in this case is complex, and it can be verified whether the generated model can describe the interaction between the objects in the system well. The case of LibMS describes the need for students to borrow and return books. In this case, the system operation is complex, which can verify whether the generated design model can encapsulate the system operation well. The case of LoanPS describes the user's demand for loans. To fulfill this demand, it is necessary to call the services of the third party, which can verify whether the generated design model can well describe the scenario where the

system interacts with other systems to complete functions. The ParkShare case allows people to rent or share their available parking spaces, which are relatively small in scale. The AirMS case addresses the needs of airport equipment management and has a larger scale. Both case studies are derived from real systems and have requirement models constructed by experienced developers. The complexity of the six requirement models is close to the complexity of the enterprise monolithic system that this study is aimed at, and it is suitable for the data set of this experiment. Make statistics on the number of model elements in the requirement model, and the statistical results are shown in Table 4.

Table 4 The Complexity of Requirements Models

Case Study	Actor	Use Case	SO	OCL	Entity Class	Association
ATM	2	6	15	65	3	4
CoCoME	3	16	43	198	13	20
LibMS	7	19	45	303	11	17
LoanPS	5	10	34	127	12	8
ParkShare	4	7	12	48	7	11
AirMS	6	33	58	336	15	19
Sum	27	91	207	1077	61	79

* SO is the abbreviations of system operations. OCL refers to the number of OCL statements that can be converted into messages in the sequence diagram.

5.4 Evaluation Metrics

5.4.1 Transformation Rule Metrics

These metrics evaluate the extent to which the proposed transformation rules cover the transformation from requirements models to design models.

- **Success rate of entity-to-class transformation.** $CSRate = RC/AC$, where RC denotes the number of correct entity classes in the class diagram, and AC denotes the number of entities in the conceptual class diagram.
- **Success rate of association-to-relationship transformation.** $ASRate = RA/AA$, where RA denotes the number of correct relationships between entity classes in the class diagram, and AA denotes the number of associations in the conceptual class diagram.
- **Correctness of encapsulation of operations.** $OSRate = RO/AO$, where RO indicates the number of system operations in the class diagram that are correctly encapsulated into interface classes, and AO indicates the total number of system operations in contracts.
- **Success rate of message generation in sequence diagram.** $MSRate = RM/AM$, where AM refers to the number of OCL expressions to be converted, and RM refers to the number of messages correctly converted from OCL expressions.

769 Similarly, we can calculate $Def - MSRate$ and $Post - MSRate$ for the contract's
770 definition and postcondition parts, respectively.

771 5.4.2 Design Quality Metrics

772 We evaluate design quality using high-level quality attribute metrics in QMOOD. We
773 take the quality attributes of design models manually constructed by experts as the
774 basis and normalize the quality attribute indicators of automatic and semi-automatic
775 generated design models with a few fast iterations for comparison.

776 For the sake of descriptive clarity, formulas for calculating the high-level design
777 metrics in QMOOD are provided.

$$Reusability = -0.25 * DCC + 0.25 * CAM + 0.5 * CIS + 0.5 * DSC$$

778

$$Flexibility = 0.25 * DAM - 0.25 * DCC + 0.5 * MOA + 0.5 * NOP$$

779

$$Understandability = -0.33 * ANA + 0.33 * DAM - 0.33 * DCC \\ + 0.33 * CAM - 0.33 * NOP - 0.33 * NOM - 0.33 * DSC$$

$$Functionality = 0.12 * CAM + 0.22 * NOP + 0.22 * CIS + 0.22 * DSC + 0.22 * NOH$$

780

$$Extendibility = 0.5 * ANA - 0.5 * DCC + 0.5 * MFA + 0.5 * NOP$$

781

$$Effectiveness = 0.2 * ANA + 0.2 * DAM + 0.2 * MOA + 0.2 * MFA + 0.2 * NOP$$

782 The meanings of the variables on the right-hand side of the equation are shown in the
783 following Table 5.

784 5.5 Evaluation Results

785 5.5.1 RQ1: To what extent do the transformation rules proposed 786 by our approach cover various situations

787 Our tool can successfully and correctly generate all class diagrams of these six case
788 studies. The results of the sequence diagram generation are shown in Table 6. On
789 average, 98.50% (DSRate) of the expressions in the definition part of the contract
790 is correctly transformed into messages in the sequence diagram, 93.96% (PSRate) in
791 the post-condition part, and 94.80% (MSRate) of the overall messages are correctly
792 generated. This result shows that our proposed rule can completely cover the case

Table 5 Meaning of Variables

Variables	Meaning
DSC	Design Size in Classes
NOH	Number of Hierarchies
ANA	Average Number of Ancestors
DAM	Data Access Metric
MFM	Measure of Functional Modularity
DCC	Direct Class Coupling
CAM	Cohesion Among Methods in Class
MOA	Measure of Aggregation
MFA	Measure of Function Abstraction
NOP	Number of Polymorphic Methods
CIS	Class Interface Size
NOM	Number of Methods

Table 6 The Generation Result of Sequence Diagram Message

Case Study	Def-OCL	Def-Message	DSRate	Post-OCL	Post-Message	PSRate	MSRate
ATM	9	9	100.00%	56	48	85.71%	87.69%
CoCoME	35	34	97.14%	163	156	95.71%	95.71%
LibMS	59	57	96.61%	244	223	91.39%	92.41%
LoanPS	19	19	100.00%	108	104	96.30%	96.85%
ParkShare	5	5	100.00%	43	42	97.67%	97.92%
AirMS	73	73	100.00 %	263	251	95.44%	96.43%
Sum	200	197	98.50%	877	824	93.96%	94.80%

* Def-OCL refers to the OCL expressions to be converted in the definition part of the system operation contract, and Def-Message refers to the number of messages correctly converted from Def-OCL; Post-OCL refers to the OCL expressions to be converted in the post-condition part of the system operation contract, and Post-Message refers to the number of messages correctly converted from Post-OCL.

of class diagram generation and can basically cover the case of sequence diagram generation.

After analysis, the failure to generate messages includes the following:

1. The developed tool cannot handle multiple nested If-else statements.
2. The contract statement is wrongly written.
3. The return value of the contract is not a variable, but an OCL operation statement.

5.5.2 RQ2: How is the quality of the generated design models, and how does it compare to manually created models

In our study, we used the design model that conforms to the anemic domain model constructed by experts as the correct reference model. Based on this reference model, we calculated the model quality metrics for design models manually constructed by graduate students in the software engineering field and those automatically generated by our tool, as shown in Table 7. The arithmetic mean calculation based on six experimental datasets reveals that design models generated by RapidOOD achieve 89.20% (without human intervention) and 94.56% (under human-in-the-loop collaboration) of the quality assessment metrics attained by expert-developed models, respectively.

Table 7 Comparison of Results: Pure MDE, LLM-Only, and RapidOOD

Case Study	Quality	LLM Only		Pure MDE	RapidOOD(LLM+Rule)	
		Single LLM	Multi LLM	Rule Only	Without Human	Human in the Loop
ATM	Reusability	0.49	0.90	0.58	0.87	1
	Flexibility	0.15	0.24	0.43	0.8	1
	Understandability	-0.21	-1.16	-0.56	-0.93	-1
	Funcationality	0.44	0.49	0.65	0.98	0.98
	Extendibility	0.50	0.42	0.36	1	1
	Effectiveness	0.32	0.34	0.80	0.99	0.99
Case Study	Quality	Single LLM	Multi LLM	Rule Only	Without Human	Human in the Loop
CoCoME	Reusability	0.38	0.47	0.68	0.86	1
	Flexibility	0.18	0.23	0.08	0.98	1
	Understandability	-0.02	-0.15	-0.67	-0.98	-1
	Funcationality	0.37	0.29	0.69	0.96	0.96
	Extendibility	0.25	0.47	0.45	0.86	1
	Effectiveness	0.39	0.37	0.26	0.93	0.98
Case Study	Quality	Single LLM	Multi LLM	Rule Only	Without Human	Human in the Loop
LibraryMS	Reusability	0.34	0.42	0.56	0.90	1
	Flexibility	0.32	0.37	0.77	0.98	1
	Understandability	-0.24	-0.12	-0.56	-0.97	-1
	Funcationality	0.35	0.30	0.59	0.99	0.99
	Extendibility	0.32	0.33	0.36	0.89	0.97
	Effectiveness	0.36	0.43	0.60	0.96	0.98
Case Study	Quality	Single LLM	Multi LLM	Rule Only	Without Human	Human in the Loop
LoanPS	Reusability	0.51	0.56	0.16	0.89	0.98
	Flexibility	0.21	0.08	0.36	0.96	1
	Understandability	-0.42	-0.52	-0.17	-0.98	-1
	Funcationality	0.39	0.48	0.23	0.99	0.99
	Extendibility	0.42	0.48	0.32	0.94	1
	Effectiveness	0.34	0.26	0.40	0.97	1
Case Study	Quality	Single LLM	Multi LLM	Rule Only	Without Human	Human in the Loop
ParkShare	Reusability	0.32	0.66	0.35	0.96	1
	Flexibility	0.01	0.14	0.54	0.92	1
	Understandability	-0.05	-0.21	-0.35	-0.97	-1
	Funcationality	0.22	0.38	0.42	0.98	0.98
	Extendibility	0.50	0.48	0.32	0.93	1
	Effectiveness	0.20	0.30	0.57	0.97	0.99
Case Study	Quality	Single LLM	Multi LLM	Rule Only	Without Human	Human in the Loop
AirMS	Reusability	0.55	0.28	0.19	0.87	0.99
	Flexibility	0.23	0.20	0.24	0.90	0.97
	Understandability	-0.32	-0.17	-0.20	-0.93	-0.99
	Funcationality	0.64	0.36	0.24	0.93	0.96
	Extendibility	0.39	0.44	0.44	0.95	0.96
	Effectiveness	0.33	0.34	0.50	0.95	0.96

¹ We adopt the design model manually constructed by experts based on the requirements model as the benchmark. The results in the table are the ratios of quality attributes compared to the benchmark.

² *LLM Only* column: results obtained solely by posing questions to the LLM. *Single LLM*: Requirements model provided to LLM for direct design model generation. *Multi LLM*: Design model from 5 "Generate - Evaluate - Optimize" iteration with 3 LLM nodes.

³ *Pure MDE* column: results obtained by rule-based design generation and rule-based refactoring only, without LLM involvement.

⁴ *RapidOOD* column: *Without Human* - initial design by Design Generation Agent only; *Human in the Loop* - initial design adjusted by human per evaluation and refactoring suggestions after a few iterations.

In all LLM-involved experiments, we used **GPT-4o** (**gpt-4o**) as the underlying language model, which is hardcoded in the tool implementation. Each LLM call follows a two-part prompt structure: a *format-constrained System Prompt* paired with a *structured User Prompt*. The System Prompt is responsible for enforcing output format constraints (e.g., requiring the model to respond exclusively in well-formed JSON or UML-compatible notation), while the User Prompt supplies the contextual input data—including class diagrams, OCL contracts, and few-shot examples—together with an explicit specification of the expected output format. This separation of concerns

ensures that format compliance is handled at the system level, keeping User Prompts focused on domain-specific content.

We also compared the performance of our method with that of directly using large language models (LLMs), including single LLM and multi LLMs. For single LLMs, we provided the requirement model to the LLM and requested it to directly generate the corresponding design model. As for multi LLMs, we created a workflow that includes three LLM nodes. One was responsible for design generation, another for quality assessment, and the third for refactoring. The process of generation-assessment-refactoring was iterated five times, after which the final result was output.

In addition, we compared our method against *Pure MDE*, which performs rule-based design generation and rule-based refactoring exclusively, without any LLM involvement. This baseline represents a traditional model-driven engineering pipeline where transformation rules fully determine the output. As shown in Table 7, Pure MDE achieves an average quality score of approximately 43.5% relative to the expert-constructed reference models across the six case studies. Although this result surpasses the Single LLM approach (approximately 32.5% on average), it falls substantially short of RapidOOD, which achieves 89.20% (without human intervention) and 94.56% (human-in-the-loop).

The comparison protocol therefore includes five groups: Single LLM, Multi-LLM, Pure MDE/Rule Only, RapidOOD without Human, and RapidOOD with Human. The first two groups test LLM-only generation and iterative LLM generation, respectively; Pure MDE tests deterministic model transformation; and the two RapidOOD groups test the contribution of the integrated rule-LLM workflow and human collaboration.

According to the findings, we can conclude that:

1. The initial designs generated by our method can achieve a quality level close to that of experts. Moreover, with the assistance of software, after a few manual adjustments, the designs can reach the level of experts in some indicators, and overall, attain the same quality standard as that of experts.
2. Compared with methods that merely rely on LLMs, our approach exhibits substantial advantages in quality metrics. Through analysis, the primary reason is that the scale of results output by LLMs is limited, and this limited scale confers better understandability. In terms of indicators such as Reusability, Flexibility, Functionality, Extendibility, and Effectiveness, methods solely using LLMs can hardly reach a level comparable to ours. This reflects the benefits of integrating the model-driven approach with LLMs. Employing a rule-based method to generate the initial design can: 1) Ensure the initial scale of the design model; 2) Guarantee the traceability between the design model and the requirement model; 3) Prevent the introduction of hallucinations from the output of LLMs.
3. Compared with the Pure MDE baseline, RapidOOD achieves substantially higher quality scores across the six case studies. While Pure MDE guarantees structural completeness and full traceability through deterministic rule application, it lacks the capacity for semantic-level optimization and context-aware design decisions.

5.5.3 RQ3: Does our automated approach for constructing design models offer a time efficiency advantage over manual approaches

In the process of manually constructing the design model, we provided the requirement model, which typically includes use case diagrams, class diagrams, and system sequence diagrams. In our approach, we additionally employed OCL contracts to constrain system operations. The time spent on writing OCL contracts constitutes the primary time consumption of our automated method. Therefore, we compared the time consumed by OCL contracts with that of manually constructing design models, as shown in Table 8. As can be seen, the average time taken to construct OCL contracts is 4.6 hours, while the average time for manually constructing design models is 6.4 hours. Our approach can save 28.1% of the time consumption, achieving 1.39 times the efficiency of the manual method.

Table 8 Time Consuming of Automated Mean and Manual Mean

Case Study	OCL Cost(h)	Manual Mean(h)	Time Saving Rate
ATM	1.3	2.0	35.00%
CoCoME	4.9	7.5	34.67%
LibMS	6.4	8.3	22.89%
LoanPS	6.9	7.2	4.17%
ParkShare	3.1	4.2	26.19%
AirMS	5.0	9.2	45.65%
Average	4.6	6.4	28.09%

5.6 Discussion and Limitation

Using lightweight formal requirement models as input, our approach can successfully generate design models that conform to the anemic domain model. Our approach outperforms graduate students in the software engineering field in terms of completeness and consistency with expert-constructed design models. However, our method has the following limitations.

Our approach does not currently support the handling of multiple nested logical judgments. This limitation restricts the ability of our tool to capture complex relationships and constraints within the requirements. Nested logical judgments often occur in real-world systems, especially in those with intricate business rules. Therefore, extending our approach to support such constructs would significantly enhance its applicability and robustness.

Our method relies on users to correctly write OCL (Object Constraint Language) contracts. While OCL provides a powerful means to express constraints, it is a specialized language that may not be familiar to all stakeholders involved in the software development process. Thus, the need for expertise in OCL might limit the usability of our approach, particularly for non-technical stakeholders. Providing better support for

889 OCL validation or integrating a user-friendly interface for generating OCL contracts
890 could help mitigate this issue.

891 Currently, our approach only generates anemic domain model designs and cannot
892 support the generation of a wider variety of design models, such as rich domain models
893 or domain-driven design patterns. Anemic domain models, while useful, do not always
894 capture the full richness of the domain logic. Expanding our approach to support
895 other design patterns could improve the expressiveness and flexibility of the generated
896 design models.

897 5.7 Threats to Validity

898 There are several threats to the validity of our study.

899 The dataset used in this experiment consists of only six requirement models, limit-
900 ing the generalizability of our findings. The automatic generation of the class diagram
901 in the design model is only conducted six times, and the evaluation of the success rate
902 of the automatic generation of the class diagram may not be sufficient to fully validate
903 the effectiveness of our approach across different domains and complexities.

904 The complexity of the data centralized system used in this experiment is close to
905 that of a single enterprise information system. However, there are still differences in
906 the actual business processes and requirements, which may affect the rationality of
907 the experimental evaluation. To address this, future work should include experiments
908 with a broader range of systems and varying levels of complexity.

909 We use expert-constructed design models as the correct answers for comparison.
910 While this approach is common, it may introduce human bias. Expert opinions can
911 vary, and there may not be a single correct design solution. To mitigate this threat,
912 we could conduct additional studies involving multiple experts and consensus-building
913 processes to ensure the reliability of the baseline models.

914 6 Conclusion and Future Work

915 This paper proposes the RapidOOD approach to automated design model generation
916 from lightweight formal requirements models. Work like RM2PT can help achieve a val-
917 idated requirements model by automatically generating prototypes from requirements
918 models. RapidOOD generates OO design models from this validated requirements
919 model, further alleviating the software development problem. The generated design
920 models can be used as references for designers and shorten the time-consuming design
921 phase. Six case studies evaluated the capabilities of RapidOOD. The experiment result
922 is satisfactory and shows that the approach is effective and efficient.

923 In the future, We will consider combining the anemic model and domain model
924 design philosophies to guide the automated generation of designs, thus addressing the
925 automatic generation of design models for a broader range of systems.

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